

THE PART I FOUND ACTUALLY CONFUSING

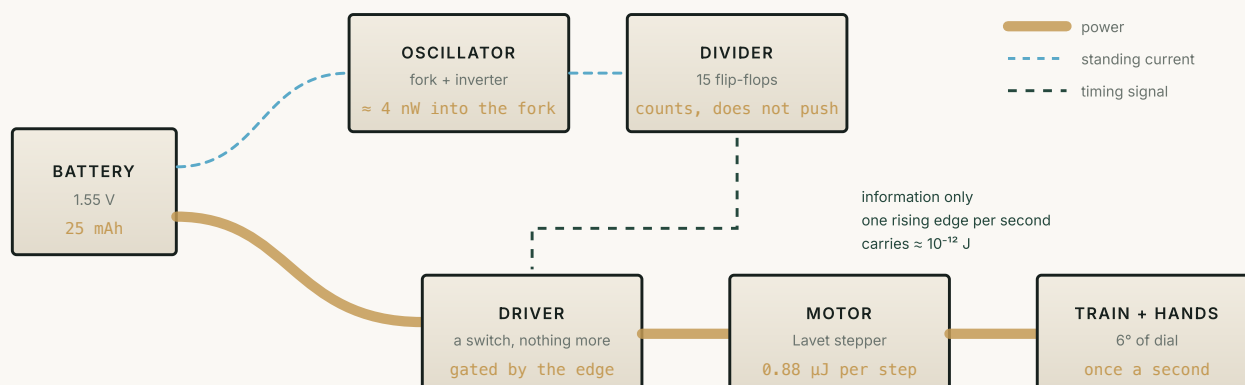
Electricity → quartz → motor

The block diagram makes it look like a relay race, with something being handed from the crystal to the motor. It is not. Almost nothing crosses that gap. I have drawn the chain again here, at the two places where energy genuinely changes form and the one place where it deliberately does not.

00 / THE CORRECTION

The crystal never drives anything

There are two separate paths out of the battery. A thin one keeps the crystal ringing, at about 4 nanowatts. A fat one drives the motor, at about 0.9 microjoules per second. Between them runs a third line that carries no useful power at all: a logic edge, once a second, saying *now*.



The battery does all the work in both branches. The crystal's job is to decide *when* the fat branch is allowed to fire, and it exerts that decision through a gate voltage, not through a force. Energy stored in the vibrating fork is about 1 nJ, roughly one nine-hundredth of a single motor pulse. Even if you could tap it, it could not move a hand.

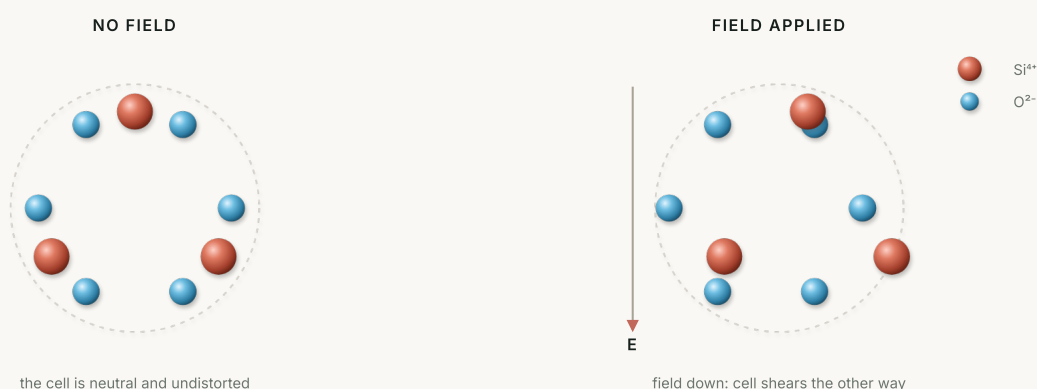
The one thing I would keep from all of this: the crystal is a clock, not an engine.

Nothing mechanical propagates from the crystal to the hands. The two mechanical systems in the watch, the vibrating fork and the turning rotor, are never connected to each other.

01 / FIRST TRANSDUCTION

How a voltage moves a rock

Quartz is SiO_2 in a lattice with no centre of symmetry. Push the silicon and oxygen sublattices in opposite directions with an electric field and they do not slide back symmetrically, so the unit cell changes shape. That is the converse piezoelectric effect, and its size for quartz is $d_{11} \approx 2.3 \text{ pm/V}$: two picometres of strain per volt, about a hundredth of an atomic diameter.

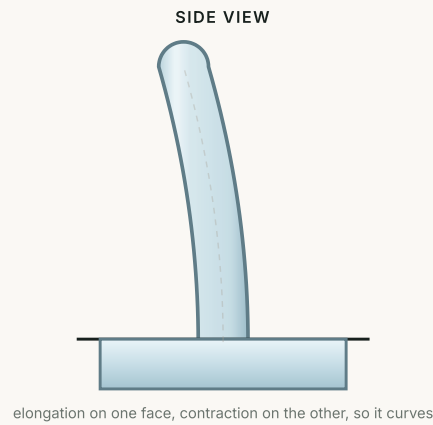
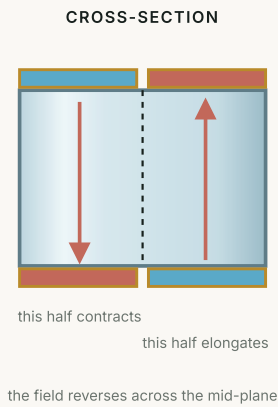


Displacements here are drawn thousands of times larger than reality. The real distortion at watch drive levels is a fraction of a picometre per cycle. On its own this is a useless amount of motion, which is why the next step matters more than this one.

02 / TURNING STRETCH INTO BENDING

Four electrodes make a lever

A tine that simply got longer would be no use. The electrodes are patterned so that the field points one way through the left half of the tine's cross-section and the opposite way through the right half. One side elongates while the other contracts, and a bar that does that has no choice but to curve.

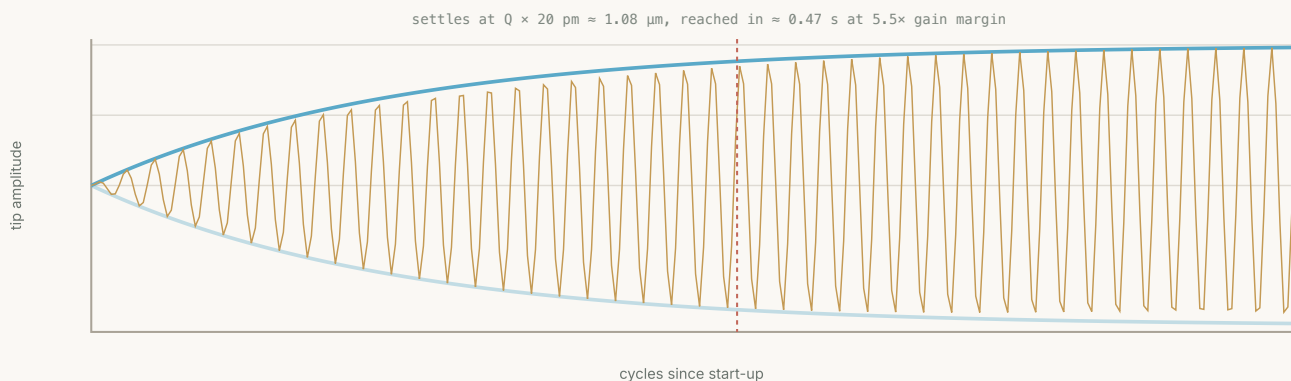


This is the whole mechanical trick of the resonator. Everything after it is resonance and counting.

03 / THE AMPLIFIER THAT IS NOT ELECTRONIC

Q turns picometres into microns

A single cycle's push is worth roughly 20 pm at the tine tip. The tip actually travels about 1 μm. The factor of fifty thousand between those numbers is not the transistor. It is Q : because the pushes arrive exactly in step with the motion, each one adds to the last, and amplitude grows until the energy fed in per cycle equals the energy lost per cycle. That balance point is reached at about Q times the single-push deflection.



Lower Q and the curve settles sooner and lower. A balance wheel sits at $Q \approx 250$, which is the same reason it is both quicker to start and two hundred times easier to knock off frequency.

The loop, stated plainly. Bending generates charge on the electrodes. The inverter amplifies that charge signal and returns it to the same electrodes with the phase that makes the tine bend further. Loss removes energy in proportion to amplitude; gain adds it at a fixed rate; the amplitude where those cross is where it lives. The crystal is simultaneously the sensor and the actuator, which is, I think, the only reason a single inverter is enough.

04 / THE GAP WHERE NOTHING PHYSICAL CROSSES

Fifteen flip-flops, one wire, one edge

This gap is the part I did not expect. Between the crystal and the motor there is only a divider chain, and a divider does not conduct force, torque, or any meaningful power. It changes the voltage on one wire from low to high, once a second. That transition is the entire output of the timekeeping half of the watch.

CRYSTAL STORED ENERGY $\approx 1\text{ nJ}$ never leaves the fork	POWER TO KEEP IT RINGING $\approx 4\text{ nW}$ replaces damping losses only
ENERGY IN ONE MOTOR PULSE $0.88\text{ }\mu\text{J}$ $\approx 900\times$ the crystal's store	ENERGY IN THE TIMING EDGE $\approx 10^{-12}\text{ J}$ charging a gate capacitance

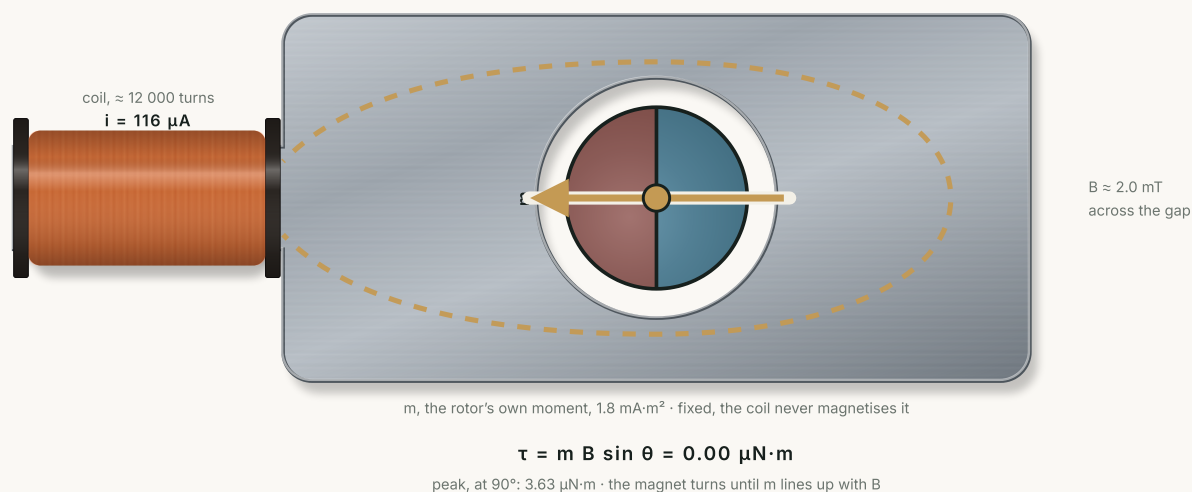
So the mental picture I would keep is not crystal → motor. It is battery → motor, with the crystal standing beside the wire holding a stopwatch.

05 / SECOND TRANSDUCTION

Current → flux → torque

Here the energy genuinely does change form, twice: electrical into magnetic, then magnetic into mechanical. The coil has roughly twelve thousand turns, so even a very small current produces a

usable magnetomotive force. The soft-iron stator is there to carry that flux around and squeeze it into the two small faces either side of the rotor.



The rotor is a permanent magnet, so it arrives with its own moment m already fixed. The coil does not magnetise the rotor; it produces a field *across* it, and a magnetic moment in an external field feels a torque proportional to the sine of the angle between them. That is the entire electromechanical conversion.

The numbers work out, as far as I can check them. The rotor is about $1.4\,\text{mm}$ across and $1.5\,\text{mm}$ long, so its volume is $2.3 \times 10^{-9}\,\text{m}^3$. SmCo with a remanence near $1\,\text{T}$ gives a magnetisation of $8 \times 10^5\,\text{A/m}$, hence a moment $m \approx 1.8\,\text{mA}\cdot\text{m}^2$. Twelve thousand turns carrying $116\,\mu\text{A}$ give 1.4 ampere-turns, which across a total air gap of a few tenths of a millimetre lands the gap field at a couple of milliteslas once reluctance is accounted for. Then $\tau = mB \approx 3.5\,\mu\text{N}\cdot\text{m}$, which is exactly the peak torque the step simulation on the lab page needs to clear the detent barrier.

06 / THIRD CONVERSION, AND THE ONLY ORDINARY ONE

Torque $\rightarrow$ half a turn $\rightarrow$ six degrees

Once there is torque on the rotor, the rest is, to my eye, unremarkable watchmaking. The rotor swings 180° , the notches in the stator bore catch it there, and a reduction train divides that half-turn down by sixty to the seconds hand and by another seven hundred and twenty to the hour hand.

WHERE ONE SECOND OF ENERGY GOES



Of the motor's 0.88 μJ , roughly a third becomes mechanical work against the detent and bearing friction.

The rest is I^2R in the coil. The fork's share of the whole budget is under half a percent.

Nothing in this ledger flows from the fork to the rotor.

07 / THE CHAIN, RESTATED

Five sentences

A voltage on patterned electrodes distorts the quartz lattice by picometres, and because the field reverses across the tine's mid-plane the distortion appears as bending rather than stretching.

Because the drive arrives in phase with the motion, roughly fifty thousand of those picometre pushes accumulate before losses catch up, so the tip ends up moving about a micron at exactly 32 768 Hz.

The bending generates charge on the same electrodes, one inverter amplifies it and sends it back, and the loop sustains itself. The frequency is set by the tine's own geometry, not by anything electrical.

Fifteen flip-flops divide that frequency to one hertz. The output is a voltage edge on one wire, carrying no meaningful energy: it opens a switch between the battery and the motor coil.

Current through twelve thousand turns produces a field of a few milliteslas across a permanent-magnet rotor, the rotor turns to line up with it, the notches in the stator catch it half a revolution later, and the train divides that down to six degrees of dial.

Educational schematic. Material constants (d_{11} , quartz E and ρ , SmCo remanence) are standard values; the currents, gap fields, and per-cycle tip increment are order-of-magnitude reconstructions I have kept consistent with the step simulation in the interactive edition. Companion to *Inside a Quartz Watch*.